Group Case Assignment 2

Marketing Analytics

Beer Market Case Study

Group B: Abdul-Rahim Baba Abdul-Mumin, Joon Kim, Likhitha Srinivas & Vivian Umelo

Q1: Our approach to brand portfolio management

Using the information from the case and the dataset (sales, price, ads, feature), there are several ways to create a balanced brand portfolio and manage a mixed marketing strategy to maximize the sales and marketing efficiency.
First, in terms of price strategy, premium brands such as Corona and Heineken should not be discounted considerably. Their strength comes from image and quality. If there is too much discount on price, there is a risk of damaging their “premium” image. It would make more sense to keep prices stable and focus on their brand image. On the other hand, mainstream brands like Coors Light and Budweiser are more price sensitive. Customers react quickly to promotions. Therefore, discounts and price cuts are likely to increase their sales volume. This matches the case where loyalty to lagers is declining, and promotions are needed to drive sales.

Second, the products that fit current consumer trends deserve more attention in terms of advertising. In other words, considerably more budget should be spent on the trendy brands to maximize the advertising ROI. Michelob Ultra is a good example. According to the case, health and sober curious trends are growing fast, and Michelob Ultra matches this demand. Focusing on Michelob Ultra on advertising could yield higher returns. Mainstream brands need to be advertised as well, but it should be done carefully and conservatively.

Third, the dataset has a variable showing if a brand was featured in the flyer. This means we can measure the display effect. Because young people are less loyal to brands, visibility in stores becomes very important. For Coors Light and Budweiser, features and displays can push sales much higher because their customers are more price-sensitive and less brand-loyal. Premium brands could also benefit, but the effect would not be as strong since their buyers usually make more intentional choices based on image and quality.

Fourth, it is important to follow the current industry trends in order to appeal to customers. Premiumization and import growth are evident, and the case shows that Modelo Especial is already outperforming Bud Light in some markets. It means that Corona and Heineken should get more shelf space. At the same time, health and low-calorie demand is growing fast as well, so Michelob Ultra and non-alcoholic beers should be expanded in the product line. This way, the supermarket could create a well-balanced, trend-following product portfolio.

Q2.
Model Equations Estimated:
Own Price Elasticity Models (Log-Linear):
$\log(SalesA) = \beta 0 + \beta 1\log(PriceA) + \beta 2\log(AdvertisingA) + \beta 3FeatA + \varepsilon$

Cross-Price Elasticity Models:
$\log(SalesA) = \beta 0 + \beta 1\log(PriceA) + \beta 2\log(PriceB) + \beta 3\log(PriceC) + \beta 4\log(PriceD) + \beta 5\log(PriceE) + \beta 6\log(AdvertisingA) + \varepsilon$

Feature Impact Models (Linear):
$Sales\_i = \beta_0 + \beta_1 \cdot Feature\_i + \varepsilon$

Results Analysis *(screenshot attached below)***:**
1. Own Price Elasticity Models (log-log):

Brand	**Price Elasticity**	**Adv Elasticity**	**Feature Effect**	**R²**	**Significance**
A (Coors Light)	-0.073	0.195	-0.002	0.7309	Price & Adv significant
B (Budweiser)	-0.091	0.143	-0.005	0.5948	Price & Adv significant
C (Corona)	-0.068	0.138	-0.0002	0.7437	Price & Adv significant
D (Michelob Ultra)	-0.036	0.164	-0.0002	0.7297	Price & Adv significant
E (Heineken)	-0.022	0.056	-0.007	0.3315	Price & Adv significant

<u>Price Elasticities</u> - All five beer brands demonstrate inelastic demand with price elasticities ranging from -0.022 to -0.091, meaning price changes have relatively modest impacts on sales volumes. Brand B (Budweiser) exhibits the highest price sensitivity at -0.091, indicating that a 1% price increase results in a 0.091% decrease in sales, while brand E (Heineken) shows the lowest price sensitivity at -0.022, reflecting its super-premium positioning and strong pricing power. Brand D (Michelob Ultra), despite its premium positioning, demonstrates surprisingly low price sensitivity at -0.036, suggesting its health-conscious differentiation creates substantial pricing flexibility.

Advertising Elasticities - Advertising elasticities across all brands are positive and statistically significant, though the magnitude varies considerably by brand. Brand A (Coors Light) demonstrates the highest advertising return on investment with an elasticity of 0.195, meaning a 1% increase in advertising spending generates a 0.195% increase in sales. Brand B (Budweiser), brand C (Corona), and brand D (Michelob Ultra) show moderate advertising effectiveness with elasticities between 0.138 and 0.164. Brand E (Heineken) exhibits the lowest advertising elasticity at 0.056, suggesting that additional advertising spending produces diminishing returns for this super-premium brand, likely because its target consumers are already well-aware of the brand and less influenced by mass advertising.

Feature effects in the log-linear models are statistically insignificant across all five brands, with p-values ranging from 0.0993 to 0.9476. This finding suggests that the simple inclusion of a feature dummy variable in the log specification does not adequately capture how promotional features influence sales, indicating potential model misspecification or that features operate through different mechanisms than captured in this framework.

2. Cross-Price Elasticity Analysis:

The cross-price elasticity analysis reveals a striking absence of statistically significant competitive substitution effects across all five beer brands. None of the models show significant cross-price elasticities between brands, with all competitor price coefficients having p-values well above conventional significance thresholds. For brand A, competitor prices from B, C, D, and E show no significant effects (p-values ranging from 0.4304 to 0.8886). Similarly, brand B sales are unaffected by competitor pricing, with all cross-price coefficients statistically insignificant (p-values from 0.3780 to 0.9732). C, D, and E exhibit the same pattern, with no evidence of price-driven consumer switching between brands.

This lack of cross-price sensitivity indicates strong brand differentiation in the market, where each brand occupies a distinct positioning that limits direct price competition. Consumers appear to exhibit strong brand preferences and loyalty, choosing their preferred brand based on factors beyond price. This finding suggests that competitive pricing responses may be less critical than maintaining brand equity and differentiation.

3. Linear Feature Effects:

Brand	**Feature Coefficient**	**P-value**	**Interpretation**
A (Coors Light)	-1.28	0.0791	Not significant
B (Budweiser)	-0.79	0.0961	Not significant
C (Corona)	+0.67	0.1234	Not significant

D (Michelob Ultra)	+0.05	0.9174	Not significant
E (Heineken)	-0.73	0.0381	Significant (negative)

The linear feature models reveal that promotional features have no statistically significant positive effect on sales for any brand. Coors Light, Budweiser, Corona, and Michelob Ultra all show insignificant feature coefficients with p-values above 0.05. Most surprisingly, Heineken demonstrates a statistically significant negative feature effect of -0.73 units (p=0.0381), suggesting that featuring this super-premium brand in promotional materials actually reduces sales, likely because discounting damages the brand's premium image and equity.

Marketing Mix Strategy

Pricing: Keep prices stable or reduce slightly for Brand B (Budweiser) given its high price sensitivity. Brand A (Coors Light) and Brand C (Corona) have moderate sensitivity, allowing limited tactical changes when necessary. Brand D (Michelob Ultra) and Brand E (Heineken) show very low price elasticities, so both can sustain premium pricing without risking major sales losses.

Advertising: Increase advertising spend for Brand A (Coors Light), which delivers the highest return on ad investment. Support Brand D (Michelob Ultra) as the second most responsive brand to advertising, using campaigns to strengthen its health-focused positioning. Maintain consistent spend for Brand B (Budweiser) and Brand C (Corona) to protect share, while Brand E (Heineken) should focus only on targeted, premium placements rather than broad advertising.

Features/Promos: Eliminate feature promotions for Brand E (Heineken), since discounting hurts its premium image and reduces sales. For Brands A, B, C, and D, feature effects are not significant, so promotions should be used cautiously and only when combined with stronger tactics (price cuts, displays) to avoid wasted spend.

Competitive Response: No significant cross-price effects were found across the portfolio. This means all brands can pursue independent strategies without reacting to competitor pricing moves, focusing instead on brand equity and positioning to drive sales.

Q3: Model Improvements

- No interaction effects between marketing variables: The current models assume price, advertising, and features operate independently, but in practice these marketing mix elements likely interact synergistically or antagonistically.
  Interaction Effects Improved Model Equation;

$\log(SalesA) = \beta 0 + \beta 1\log(PriceA) + \beta 2\log(AdvertisingA) + \beta 3FeatureA + \beta 4(\log(PriceA)\times FeatureA) + \beta 5(\log(AdvertisingA)\times FeatureA) + \epsilon$

What's Needed: The estimation requires creating interaction terms through data transformation in SAS, specifically multiplying the logged price by logged advertising variables, the logged price by the feature dummy, and logged advertising by the feature dummy. These interaction variables would then be added as additional regressors in the PROC REG procedure alongside the main effects.

- No Carryover effects: The models treat each time period as independent, ignoring that advertising builds awareness over multiple periods and that past sales influence current demand through habit formation and inventory effects.
  Advertising Carryover (Adstock) Model Equation;
  $\log(Sales\_i,t) = \beta_0 + \beta_1 \cdot \log(Price\_i,t) + \beta_2 \cdot Adstock\_i,t + \beta_3 \cdot Feature\_i,t + \varepsilon\_t$
  where: $Adstock\_i,t = Advertising\_i,t + \lambda \cdot Adstock\_i,t-1$ (with $0 < \lambda < 1$)

  What's Needed: Estimation requires non-linear optimization techniques available through PROC MODEL in SAS because the decay parameter λ must be estimated simultaneously with the other coefficients rather than specified a priori. Researchers would need to create the lagged adstock variable recursively, where each period's adstock equals current advertising plus λ times the previous period's adstock value. Alternative approaches include testing multiple fixed values of λ (such as 0.3, 0.5, 0.7) and selecting the specification that maximizes R-squared or minimizes information criteria.

- Time effects ignored: Over the 1,000-day observation period, market conditions, competitive dynamics, consumer preferences, and brand health may have evolved, but the models assume stable coefficients throughout.
  Time Effects Model Equation;
  $\log(Sales\_i,t) = \beta_0 + \beta_1 \cdot \log(Price\_i,t) + \beta_2 \cdot \log(Advertising\_i,t) + \beta_3 \cdot Feature\_i,t + \beta_4 \cdot \log(Sales\_i,t-1) + \beta_5 \cdot Time\_t + \beta_6 \cdot Time^2\_t + \alpha\_i + \varepsilon\_t$
  where α_i = brand-specific fixed effect

  What's Needed: Estimation requires creating a lagged sales variable using the LAG function in a SAS data step, which shifts the sales values back one period for each brand separately to avoid mixing observations across brands. Time trend variables can be generated from the existing Day variable, creating both linear (Day) and quadratic (Day squared) terms.

Own Price Elasticities (log-log)

The SAS System

The REG Procedure
Model: loglogBrandA
Dependent Variable: logsalesA

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	7.38007	2.46002	901.59	<.0001
Error	996	2.71762	0.00273		
Corrected Total	999	10.09769			

Root MSE	0.05224	R-Square	0.7309
Dependent Mean	4.66112	Adj R-Sq	0.7301
Coeff Var	1.12066		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.04158	0.01393	290.08	<.0001
logPriceA	1	-0.07264	0.00913	-7.96	<.0001
logAdvertisingA	1	0.19537	0.00382	51.16	<.0001
FeatA	1	-0.00193	0.00360	-0.54	0.5926

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The SAS System

The REG Procedure
Model: loglogBrandB
Dependent Variable: logsalesB

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3.49588	1.16529	487.39	<.0001
Error	996	2.38135	0.00239		
Corrected Total	999	5.87723			

Root MSE	0.04890	R-Square	0.5948
Dependent Mean	4.55711	Adj R-Sq	0.5936
Coeff Var	1.07298		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.17009	0.01361	306.36	<.0001
logPriceB	1	-0.09113	0.00749	-12.17	<.0001
logAdvertisingB	1	0.14317	0.00392	36.51	<.0001
FeatB	1	-0.00522	0.00316	-1.65	0.0993

The SAS System

The REG Procedure
Model: loglogBrandC
Dependent Variable: logsalesC

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3.27053	1.09018	963.58	<.0001
Error	996	1.12686	0.00113		
Corrected Total	999	4.39739			

Root MSE	0.03364	R-Square	0.7437
Dependent Mean	4.54949	Adj R-Sq	0.7430
Coeff Var	0.73934		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.14335	0.00950	436.04	<.0001
logPriceC	1	-0.06784	0.00546	-12.44	<.0001
logAdvertisingC	1	0.13824	0.00263	52.48	<.0001
FeatC	1	-0.00015350	0.00234	-0.07	0.9476

The SAS System

The REG Procedure
Model: loglogBrandD
Dependent Variable: logsalesD

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3.76015	1.25338	896.20	<.0001
Error	996	1.39295	0.00140		
Corrected Total	999	5.15310			

Root MSE	0.03740	R-Square	0.7297
Dependent Mean	4.59416	Adj R-Sq	0.7289
Coeff Var	0.81401		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.12584	0.01053	391.71	<.0001
logPriceD	1	-0.03600	0.00590	-6.10	<.0001
logAdvertisingD	1	0.16427	0.00319	51.54	<.0001
FeatD	1	-0.00022064	0.00271	-0.08	0.9350

The SAS System

The REG Procedure
Model: loglogBrandE
Dependent Variable: logsalesE

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	0.46279	0.15426	164.61	<.0001
Error	996	0.93340	0.00093715		
Corrected Total	999	1.39619			

Root MSE	0.03061	R-Square	0.3315
Dependent Mean	4.45417	Adj R-Sq	0.3295
Coeff Var	0.68729		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.31008	0.00863	499.17	<.0001
logPriceE	1	-0.02161	0.00500	-4.32	<.0001
logAdvertisingE	1	0.05627	0.00264	21.28	<.0001
FeatE	1	-0.00710	0.00334	-2.13	0.0336

Cross-Price Elasticities

The REG Procedure
Model: CrossBrandA
Dependent Variable: logsalesA

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	7.39101	1.23184	451.92	<.0001
Error	993	2.70668	0.00273		
Corrected Total	999	10.09769			

Root MSE	0.05221	R-Square	0.7320
Dependent Mean	4.66112	Adj R-Sq	0.7303
Coeff Var	1.12009		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.04247	0.01543	262.00	<.0001
logPriceA	1	-0.06884	0.01503	-4.58	<.0001
logPriceB	1	0.00452	0.01024	0.44	0.6589
logPriceC	1	0.00838	0.01062	0.79	0.4304
logPriceD	1	-0.02019	0.01030	-1.96	0.0502
logPriceE	1	0.00149	0.01061	0.14	0.8886
logAdvertisingA	1	0.19545	0.00382	51.19	<.0001

The REG Procedure
Model: CrossBrandB
Dependent Variable: logsalesB

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	3.49191	0.58198	242.28	<.0001
Error	993	2.38533	0.00240		
Corrected Total	999	5.87723			

Root MSE	0.04901	R-Square	0.5941
Dependent Mean	4.55711	Adj R-Sq	0.5917
Coeff Var	1.07550		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.16526	0.01505	276.78	<.0001
logPriceB	1	-0.09407	0.00960	-9.80	<.0001
logPriceA	1	0.00047536	0.01412	0.03	0.9732
logPriceC	1	-0.00466	0.00997	-0.47	0.6404
logPriceD	1	0.00853	0.00967	0.88	0.3780
logPriceE	1	0.00136	0.00996	0.14	0.8918
logAdvertisingB	1	0.14312	0.00394	36.34	<.0001

The REG Procedure
Model: CrossBrandC
Dependent Variable: logsalesC

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	3.27382	0.54564	482.23	<.0001
Error	993	1.12357	0.00113		
Corrected Total	999	4.39739			

Root MSE	0.03364	R-Square	0.7445
Dependent Mean	4.54949	Adj R-Sq	0.7429
Coeff Var	0.73937		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.14107	0.01024	404.47	<.0001
logPriceC	1	-0.06369	0.00684	-9.31	<.0001
logPriceA	1	-0.01296	0.00968	-1.34	0.1808
logPriceB	1	0.00112	0.00659	0.17	0.8645
logPriceD	1	-0.00185	0.00664	-0.28	0.7808
logPriceE	1	0.00778	0.00683	1.14	0.2554
logAdvertisingC	1	0.13823	0.00263	52.51	<.0001

The REG Procedure
Model: CrossBrandD
Dependent Variable: logsalesD

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	3.76845	0.62807	450.42	<.0001
Error	993	1.38465	0.00139		
Corrected Total	999	5.15310			

Root MSE	0.03734	R-Square	0.7313
Dependent Mean	4.59416	Adj R-Sq	0.7297
Coeff Var	0.81281		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.12764	0.01152	358.38	<.0001
logPriceD	1	-0.03288	0.00737	-4.46	<.0001
logPriceA	1	-0.00538	0.01075	-0.50	0.6169
logPriceB	1	0.01367	0.00731	1.87	0.0619
logPriceC	1	-0.00431	0.00759	-0.57	0.5701
logPriceE	1	-0.01073	0.00759	-1.41	0.1577
logAdvertisingD	1	0.16405	0.00319	51.43	<.0001

The REG Procedure
Model: CrossBrandE
Dependent Variable: logsalesE

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	0.45951	0.07659	81.19	<.0001
Error	993	0.93667	0.00094328		
Corrected Total	999	1.39619			

Root MSE	0.03071	R-Square	0.3291
Dependent Mean	4.45417	Adj R-Sq	0.3251
Coeff Var	0.68953		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	4.31012	0.00949	454.18	<.0001
logPriceE	1	-0.01950	0.00625	-3.12	0.0018
logPriceA	1	-0.00157	0.00884	-0.18	0.8588
logPriceB	1	0.00328	0.00601	0.54	0.5860
logPriceC	1	-0.00113	0.00624	-0.18	0.8560
logPriceD	1	-0.00453	0.00606	-0.75	0.4549
logAdvertisingE	1	0.05635	0.00266	21.22	<.0001

Feature Effect

The SAS System

The REG Procedure
Model: BrandA
Dependent Variable: salesA

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	346.06019	346.06019	3.09	0.0791
Error	998	111808	112.03221		
Corrected Total	999	112154			

Root MSE	10.58453	R-Square	0.0031
Dependent Mean	106.28600	Adj R-Sq	0.0021
Coeff Var	9.95853		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	107.17822	0.60807	176.26	<.0001
FeatA	1	-1.28008	0.72834	-1.76	0.0791

The SAS System

The REG Procedure
Model: BrandB
Dependent Variable: SalesB

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	147.58006	147.58006	2.77	0.0961
Error	998	53097	53.20326		
Corrected Total	999	53244			

Root MSE	7.29406	R-Square	0.0028
Dependent Mean	95.58700	Adj R-Sq	0.0018
Coeff Var	7.63080		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	96.06045	0.36608	262.40	<.0001
FeatB	1	-0.78516	0.47143	-1.67	0.0961

The SAS System

The REG Procedure
Model: BrandC
Dependent Variable: SalesC

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	93.38242	93.38242	2.38	0.1234
Error	998	39193	39.27190		
Corrected Total	999	39287			

Root MSE	6.26673	R-Square	0.0024
Dependent Mean	94.79200	Adj R-Sq	0.0014
Coeff Var	6.61103		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	94.59433	0.23602	400.79	<.0001
FeatC	1	0.67008	0.43455	1.54	0.1234

The SAS System

The REG Procedure
Model: BrandD
Dependent Variable: SalesD

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	0.53814	0.53814	0.01	0.9174
Error	998	49903	50.00319		
Corrected Total	999	49904			

Root MSE	7.07129	R-Square	0.0000
Dependent Mean	99.15900	Adj R-Sq	-0.0010
Coeff Var	7.13127		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	99.14536	0.25942	382.18	<.0001
FeatD	1	0.05309	0.51173	0.10	0.9174

The SAS System

The REG Procedure
Model: BrandE
Dependent Variable: SalesE

Number of Observations Read	1000
Number of Observations Used	1000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	44.38127	44.38127	4.31	0.0381
Error	998	10271	10.29118		
Corrected Total	999	10315			

Root MSE	3.20799	R-Square	0.0043
Dependent Mean	86.04500	Adj R-Sq	0.0033
Coeff Var	3.72827		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	86.11246	0.10652	808.42	<.0001
FeatE	1	-0.72536	0.34929	-2.08	0.0381